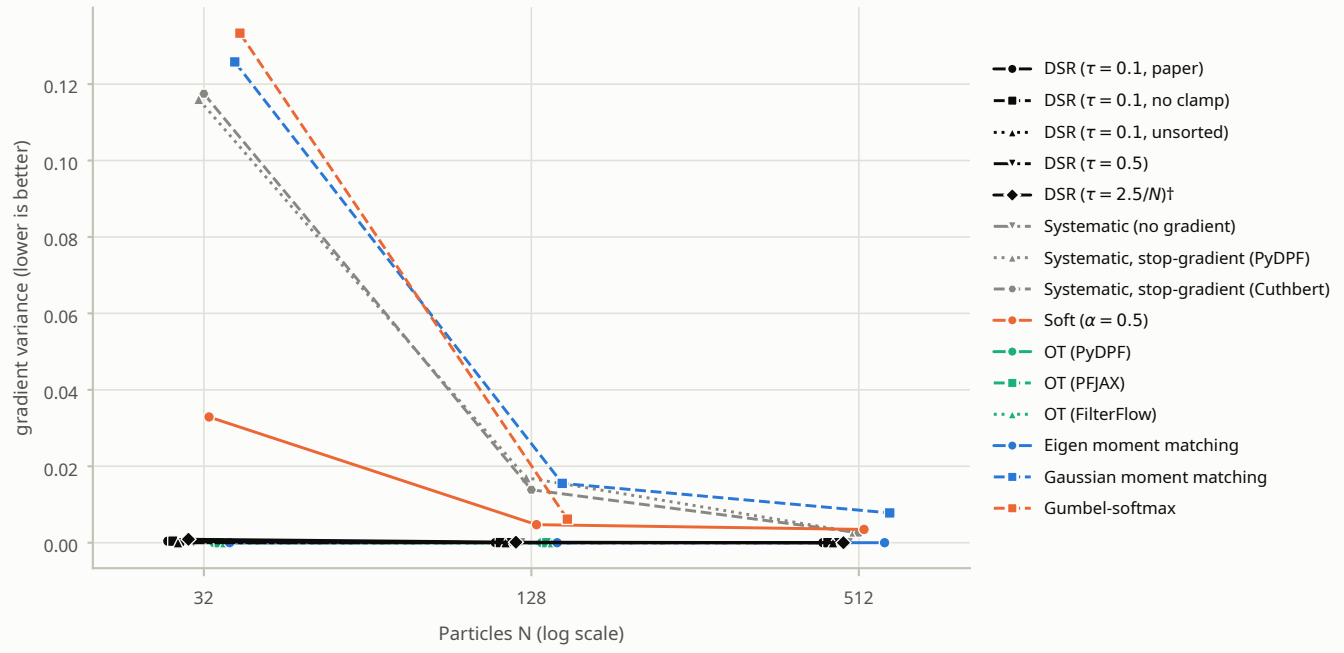
gradient variance (lower is better)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
32
128
512
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic (no gradient)
Systematic, stop-gradient (PyDPF)
Systematic, stop-gradient (Cuthbert)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax